



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCCD401

BUILDING CONSTRUCTION DRAWING

Apply advanced construction drawing

Competence

RQF Level: 4

Learning Hours



70

Credits: 7

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: BDCCD401- TVET Certificate 4 In Building Construction

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Purpose statement	This module describes the knowledge, skills and attitude required to realize Building construction drawing. It is intended to be educated to learner at TVET level IV for TVET Certificate IV in building construction. At the end of this module, the learner will be able to: Prepare materials, instrument and equipment, perform free hand architecture drawings and perform scale model.					
Leaning assumed to be in place	Safety, health and environment at workplace, fundamental mathematical analysis.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Prepare materials, instrument and equipment.	1.1.	Drawing instruments are properly collected according to the work to be done.
	1.2.	Drawing equipment are properly collected according to the work to be done.
	1.3.	Drawing materials are properly collected according to the work to be done.
2. Perform hand architecture drawings	2.1.	Work Specifications are adequately collected according to the work requirements.
	2.2.	Layout is accurately performed according to drawing sheet size.
	2.3.	Floor plan is properly drawn with hand according to the work specifications.
	2.4.	Site plan is properly drawn with hand according to the plot.
	2.5.	Foundation plan is properly drawn with hand according to the work specifications.
	2.6.	Roof plan is properly drawn according to the work specifications and the shape of floor plan.
	2.7.	Elevation views are accurately produced according to the structure shape.
	2.8.	Sections plans are accurately produced according to the position of cutting lines.
	2.9.	Location plan is properly drawn according physical plot location.
	2.10.	3D perspective is properly produced with hand according to the design.

3. Fabricate scale model.	3.1.	Materials are properly collected according to the scale model.
	3.2.	Tools are properly collected according to the scale model to be made.
	3.3.	Scale model base is properly prepared according to model size.
	3.4.	Scale model elements are properly prepared according to model size.
	3.5.	Scale model elements are properly assembled according to the structure design.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify units of measurement ✓ Identify drawing tools, equipment and materials ✓ Identify drawing signs and symbols	✓ Apply basic calculations, trigonometry and numerical formulas ✓ Apply drafting and lettering skills ✓ Use drawing tools, equipment and materials	✓ Demonstrate organisational and time management skill during design works ✓ Demonstrate persistent, team working, flexibility, punctuality at work ✓ Demonstrate the ability to Work independently with minimal supervision ✓ Pay attention to details

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare materials, instrument and equipment. 2. Perform hand architecture drawings. 3. Fabricate scale model.
Learning outcome 1: Prepare materials, instrument and equipment.	Learning hours: 10
Indicative content	

- **Materials used in technical drawings**
 - Function of drawing materials
 - Maintenance of drawing materials
 - Selection of drawing materials
- **Instrument used in technical drawings**
 - Function of drawing instruments
 - Status of drawing instruments
 - Selection of drawing instruments
- **Equipment used in technical drawings**
 - Function of drawing equipment
 - Selection of drawing equipment
 - Maintenance of drawing equipment

Resources required for the learning outcome

Equipment	- Drawing board, drawing table, drawing pins and clips.
Materials	- Eraser, drawing papers, Ink, lead.
Instrument	- T-square, set squares, French curves, protractor, ruler, ink-pen, pencils, pencils sharpener.
Facilitation techniques	• Brainstorming • Group discussion • Trainer guided • Individualized • Demonstration • Practical exercises
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions • Performance assessment • Product assessment

Learning outcome 2: Perform hand architecture drawings

Learning hours: 35

Indicative content

- **Drawing specifications**
 - Drawing instructions
 - ✓ Types of building
 - ✓ Dimensions
 - ✓ Number of rooms
 - ✓ Paper format
 - ✓ Scale
- **Drawing layout**
 - Setting of drawing paper margins
 - Placing of title block on drawing paper
 - Arrangement of architecture drawing plans
- **Drawing of floor plan**
 - Selection of drawing area
 - Drafting with pencil
 - Drawing walls to scale
 - Adding architectural features and furniture
 - Labelling of drawing units
- **Drawing site plan**
 - Identification of existing and proposed conditions
 - ✓ Landscape
 - ✓ Parking
 - ✓ Courtyard
 - ✓ Building coverage
 - ✓ Drainage system
 - ✓ Fence and boundaries
 - ✓ Walkways
 - Arrangement of site plan elements
- **Drawing of foundation plan**

➤ Identification of foundation plan elements

- ✓ Footing
- ✓ Pier
- ✓ Columns
- ✓ Pile

➤ Arrangement of foundation plan elements

• **Drawing of roof plan**

➤ Identification of roof plan elements

- ✓ Roof dimensions,
- ✓ Roof specifications for roof pitch/slope,
- ✓ Placement of vents and drainage,
- ✓ Roof material specifications

➤ Draw roof plan

• **Drawing of elevation views**

➤ Types of elevation views

- ✓ Front view
- ✓ Rear view
- ✓ Left view
- ✓ Right view
- ✓ Top view

➤ Identification of elevation view features

- ✓ Doors
- ✓ Windows
- ✓ Sill
- ✓ Lintel

➤ Draw elevations

• **Drawing of cross-sectional plans**

➤ Types of cross sections

- ✓ Longitudinal cross section
- ✓ Transversal cross section

- Perform cross sections drawing

- **Drawing of location plan**

- Identification of boundaries of the current plot to bordered plot
- Label important locations and areas
- Put in legend

- **Drawing 3D perspective**

- Realise 3D perspective
 - ✓ Sketch the Squares
 - ✓ Add the vanishing point and
 - ✓ Add orthogonal lines
 - ✓ Trace the Image

Resources required for the indicative content

Equipment	- Drawing board, drawing table, drawing pins and clips.
Materials	- Eraser, drawing papers, Ink, lead.
Instruments	- T-square, set squares, French curves, protractor, ruler, ink-pen, pencils, pencil sharpener.
Facilitation techniques	• Brainstorming • Group discussion • Trainer guided • Individualized • Demonstration • Practical exercises
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions • Performance assessment • Product assessment

Learning outcome 3: Fabricate scale model

Learning hours:25

Indicative content

- **Materials used in scale model fabrication**

- ✓ Wood
- ✓ Foam board
- ✓ Glue
- ✓ Paper
- ✓ Paint

- **Tools used in scale model fabrication**

- ✓ Craft knife
- ✓ Cutting mat
- ✓ Ruler
- ✓ Tape measure
- ✓ Pencil
- ✓ Calculator

- **Preparation of scale model base**

- Preparation of materials for scale model base.
- Sizing of scale model base

- **Preparation of scale model elements**

- Measuring parts of scale model elements
- Marking parts of scale model elements
- Cutting parts of scale model elements
- Preparing exterior materials
 - ✓ Windows
 - ✓ Door
 - ✓ Steps

- **Assembling of scale model elements**

- Constructing the solid core
- Bonding the sides
- Finishing exterior walls
- Finishing the roof

Resources required for the indicativefor the indicative content

Equipment	- Drawing board, drawing table, drawing pins and clips.
Materials	Eraser, drawing papers, Ink, lead, Wood, Foam board, Glue, Papers, Paints
Instruments	T-square, set squares, French curves, protractor, ruler, ink-pen, pencils, pencil sharpener, Paper cutting knife, brushes
Facilitation techniques	• Brainstorming • Group discussion • Trainer guided • Individualized • Demonstration • Practical exercises
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment

Integrated/Summative assessment (For specific module)

Integrated situation

Mr. KABERA, an investor who lives in Gicumbi District, Rukomo center was inspired by current lack of commercial building in the area to initiate the project of constructing a one storey commercial building that will provide commercial services for different businesses in his plot of 42m x 28m located in the above said center. When he went to the district administration to ask what is required get a construction permit to implement the project, the district permitting officer provided him the checklist of required architectural drawing items that includes: Floor plan, foundation plan, elevation views, roof plan, Sections, location plan, 3D perspectives and Scale model object which shows the general picture of the building that is planned to be built. After gathering necessary information, Mr. KABERA contacted SJAP Construction and Consultancy Company Ltd to give him a skilled technician to produce these drawing to be presented to the district one stop center. As an employee of SJAP Construction and Consultancy Company Ltd, you are requested to make these architectural drawings within seven days and put them on drawing sheet paper of A0 format, given that:

1. The shape of the building is irregular shape
2. The external length and Width of the building is 22 m to 16m respectively
3. The Thickness of the wall is 20 cm
4. The floor level height is 3.4m

5. The roof type is pitched roof with 30°
6. The scale to be used is 1/100
7. The scale for scale model is 1/40
8. The building consists of six shops at each floor
9. The dimension of Door and window are 1.4m*2.4m to 1m to 1.4m respectively
10. The window sill is at 0.7m at each floor.

Equipment	- Drawing board, drawing table, drawing pins and clips.
Materials	-Eraser, drawing papers, Ink, lead, Wood, Foam board, Glue, Papers, Paints
Instruments	- T-square, set squares, French curves, protractor, ruler, ink-pen, pencils, pencil sharpener, Paper cutting knife, brushes

Resources

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Prepare materials, instrument and equipment (20%).	1.1. Drawing instruments are properly collected according to the work to be done	T-square is selected.			1
		Set square are selected.			2
		Ruler is selected.			1
		Ink pen is selected.			2
		Pencil is prepared.			2
	1.2. Drawing equipment are properly collected according to the work to be done	Drawing board is selected.			2
		Protractor is selected.			1
		Drawing table is prepared.			1
		Drawing pins are selected.			1
		Clips are prepared selected.			2
	1.3. Drawing materials are properly collected according to the work to be done	Eraser is selected.			2
		Drawing papers format is respected.			1
		Ink is selected.			1
		lead size is selected.			1
Learning outcome 2: Perform hand architecture drawings (50%).	2.1. Work Specifications are adequately collected according to the work requirements	Dimensions is mentioned.			1
		Paper format is mentioned.			1
		Scale is mentioned.			1
		Number of rooms is mentioned.			1
		Types of building is mentioned.			1

	2.2. Layout is accurately performed according to size drawing	Drawing paper margins is set.			1
		Title block on drawing paper is placed.			1
		Architecture drawing plans are arranged.			1
	2.3. Floor plan is properly drawn with hand according to the work specifications	Drawing area is selected.			1
		Drafting with pencil is done.			1
		Walls are mentioned.			1
		Architectural features and furniture are positioned.			1
		Drawing units are labeled.			1
	2.4. Site plan is properly drawn with hand according to the plot	Landscaping is done.			1
		Parking area is mentioned.			1
		Courtyard is mentioned.			1
		Building coverage is mentioned.			1
		Drainage system is mentioned.			1
		Fence and boundaries are mentioned.			1
		Walkways is mentioned.			1
	2.5. Foundation plan is properly drawn with hand according to the work specifications	Footings are mentioned.			2
		Piers are properly mentioned.			2
		Columns are properly mentioned.			2
		Piles are properly mentioned.			2
	2.6. Roof plan is properly drawn according to the work specifications and the shape of floor plan	Roof dimensions are respected.			2
		Slope of the roof is respected.			2
		Vents and drainage elements are mentioned.			2
		Roof shape is respected.			2
	2.7. Elevation Views are accurately produced according to the work to be done	Front view elevation is produced.			1
		Rear view elevation is produced.			1
		Left view elevation is produced.			1

		Right view elevation is produced.			1
		Top view elevation is produced.			1
	2.8. Sections plans are accurately produced according to the position of cutting lines.	Longitudinal cross section is produced.			2
		Transversal cross section is produced.			2
		Units and labels are shown.			1
	2.9. Location plan is properly drawn according the plot location.	Plot boundaries are shown.			1
		Important locations and areas are labeled.			1.5
		Legends are done.			1.5
Learning outcome 3: Fabricate scale model (30%).	3.2. Materials are properly collected according to the scale model.	Wooden base is selected.			1
		Foam board is selected.			1
		Glue is selected.			1
		Paper is selected.			1
		Paint is selected.			2
	3.3. Tools are properly collected according to the scale model.	Craft knife is selected.			2
		Cutting mat is selected.			1
		Ruler is selected.			1
		Tape measure is selected.			1
		Pencil is selected.			1
		Calculator is selected.			1
	3.4. Scale model base is properly prepared according to model size.	Foam board is smoothed.			2
		Foam board is sized.			2
	3.5. Scale model elements are properly prepared according to model size.	Scale model elements are measured.			2
		Scale model elements are marked.			2
		Parts of scale model are cut.			2

	3.6. Scale model elements are properly assembled according to model size.	Solid core face is constructed.			1
		Exterior face of scale model is prepared.			2
		Junction sides are bonded.			2
		The roof elements are attached.			2
Total marks		100			
Percentage Weight age		100%			
Minimum Passing line % (Aggregate): 70%					

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